

1 Solution — GIAM 1.4

Problem 16

1.1 Problem

An ice cream shop sells 7 flavors: chocolate, vanilla, strawberry, coffee, butter pecan, mint chocolate chip, and raspberry. How many different bowls of ice cream — with three scoops — can they make?

1.2 First Fix the Assumptions

A counting problem has no single answer until two modelling questions are settled:

1. **Does order matter?** In a *bowl*, the scoops sit together and are not ranked — so order should *not* matter. (A *cone*, where one scoop sits atop another, would be different.)
2. **May a flavor repeat?** Can a bowl have two scoops of chocolate? Realistically yes; but if “bowl” is meant as a selection of three *different* flavors, then no.

Since a bowl is unordered, the answer is one of the two unordered counts below. The shop, the textbook context, and the phrasing all bear on which.

1.3 The Two Reasonable Answers

Three different flavors (no repeats). This is the direct “choose 3 of the 7 flavors” reading — a straight binomial coefficient, exactly what this section practices:

$$\binom{7}{3} = \frac{7 \cdot 6 \cdot 5}{3!} = \frac{210}{6} = \boxed{35}.$$

Scoops may repeat. Allowing {choc, choc, vanilla} and the like, this is a *multiset* count — combinations with repetition, $\binom{n+k-1}{k}$:

$$\binom{7+3-1}{3} = \binom{9}{3} = \frac{9 \cdot 8 \cdot 7}{3!} = \frac{504}{6} = \boxed{84}.$$

Given that the problem sits in the *binomial-coefficient* section, **35** is the most likely intended answer; **84** is the more physically realistic one (a real shop will happily give you two scoops of the same flavor). Both are correct *for their stated assumption*.

1.4 The Four Possible Counts at a Glance

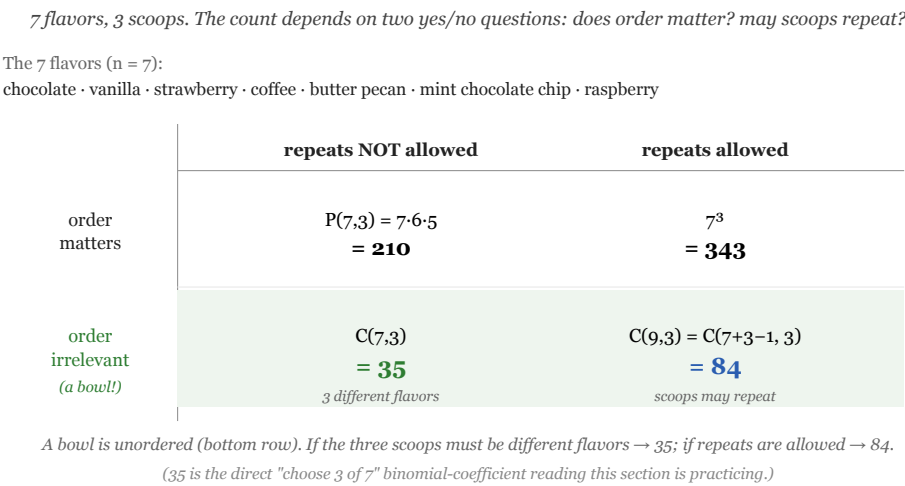


Figure 1.1: Two yes/no questions give four counts. Order matters: $P(7,3)=210$ (no repeats), $7^3=343$ (repeats). Order irrelevant — a bowl: $C(7,3)=35$ (three different flavors) or $C(9,3)=84$ (scoops may repeat). A bowl is unordered, so the answer is 35 or 84 depending on whether repeats are allowed.

	repeats not allowed	repeats allowed
order matters (a ranked stack)	$P(7,3) = 7 \cdot 6 \cdot 5 = 210$	$7^3 = 343$
order irrelevant (a bowl)	$\binom{7}{3} = 35$	$\binom{9}{3} = 84$

Table 1.1

1.5 Remark — Why a Bowl Is Unordered

The bottom row is the relevant one because a bowl has no “first” or “top” scoop — three scoops dropped in a bowl form an unordered collection. If instead the question were about a *cone* (scoops stacked in a definite order), the top row would apply: $P(7,3) = 210$ different cones of three distinct flavors, or $7^3 = 343$ if a flavor may be repeated up the stack. Recognising “bowl = unordered, cone = ordered” is the modelling step that

picks the row; it is exactly the kind of unstated assumption the textbook keeps urging us to make explicit.

1.6 Remark — Where the Multiset Count $\binom{9}{3}$ Comes From

The repeats-allowed answer uses the **stars-and-bars** idea. To build a bowl, decide *how many scoops of each of the 7 flavors* — a list of 7 non-negative counts summing to 3. Picture 3 scoops (stars) and 6 dividers (bars) separating the 7 flavor-bins:

$$\underbrace{\star\star}_{\text{choc}} \mid \mid \underbrace{\star}_{\text{straw}} \mid \mid \mid \quad \longleftrightarrow \quad 2 \text{ chocolate} + 1 \text{ strawberry}.$$

Each bowl is one arrangement of 3 stars and 6 bars in a row of 9 symbols, and choosing *which 3 of the 9 positions* are stars gives

$$\binom{3+6}{3} = \binom{9}{3} = 84.$$

In general, the number of k -scoop bowls from n flavors with repeats allowed is $\binom{n+k-1}{k}$.

1.7 Remark — How the Four Counts Relate

The four numbers 35, 84, 210, 343 are not independent; they are tied by the order/repeat toggles:

- **No repeats:** ordered $\div k!$ = unordered, i.e. $P(7, 3)/3! = 210/6 = 35 = \binom{7}{3}$. (Dividing out the 3! orderings of three *distinct* scoops.)
- **Repeats allowed:** the division by $k!$ is *not* exact, because bowls with a repeated flavor have fewer distinct orderings. That is why $343/6 = 57.2\dots$ is not an integer and *not* the answer — the multiset count 84 must be gotten by stars-and-bars, not by naive division. This subtlety (repeats break the clean “divide by $k!$ ”) is the main reason combinations-with-repetition needs its own formula.

1.8 Remark — The Real Lesson: State Your Assumptions

This problem is less about a single number than about *recognising that the wording is ambiguous*. Each of the four answers is correct under explicit assumptions:

Assumption	Count
bowl, three different flavors	35
bowl, scoops may repeat	84
cone (ordered), three different flavors	210
cone (ordered), scoops may repeat	343

Table 1.2

A careful solver writes down *which* model they are using before computing — the recurring discipline of these sections (Problems 11–13 made the same point about strict-vs-non-strict and floor-vs-truncation). For a problem in the binomial-coefficient subsection, the cleanest single answer is $\binom{7}{3} = \mathbf{35}$; for a real ice-cream shop, **84**.